

```
$ python3 day01.py
```

Python Day01

0000000000

```
>>> print("00000000000000")
```

```
00000000000000
```



```
$ cat agenda.md
```



1. Python Day01 — □□□□□□□□□□

2. □□□□

3. □□□□□□□□□□

4. □□□□□□□□

5. □□□□□□□□□□□□

6. □□

7. □□□□

8. □□□□□□

9. □□□□□□□□

10. Output

11. □□□□□□□□□□□□

12. □□□□□□□

13. f-string □□□□□□


```
int
□□
```

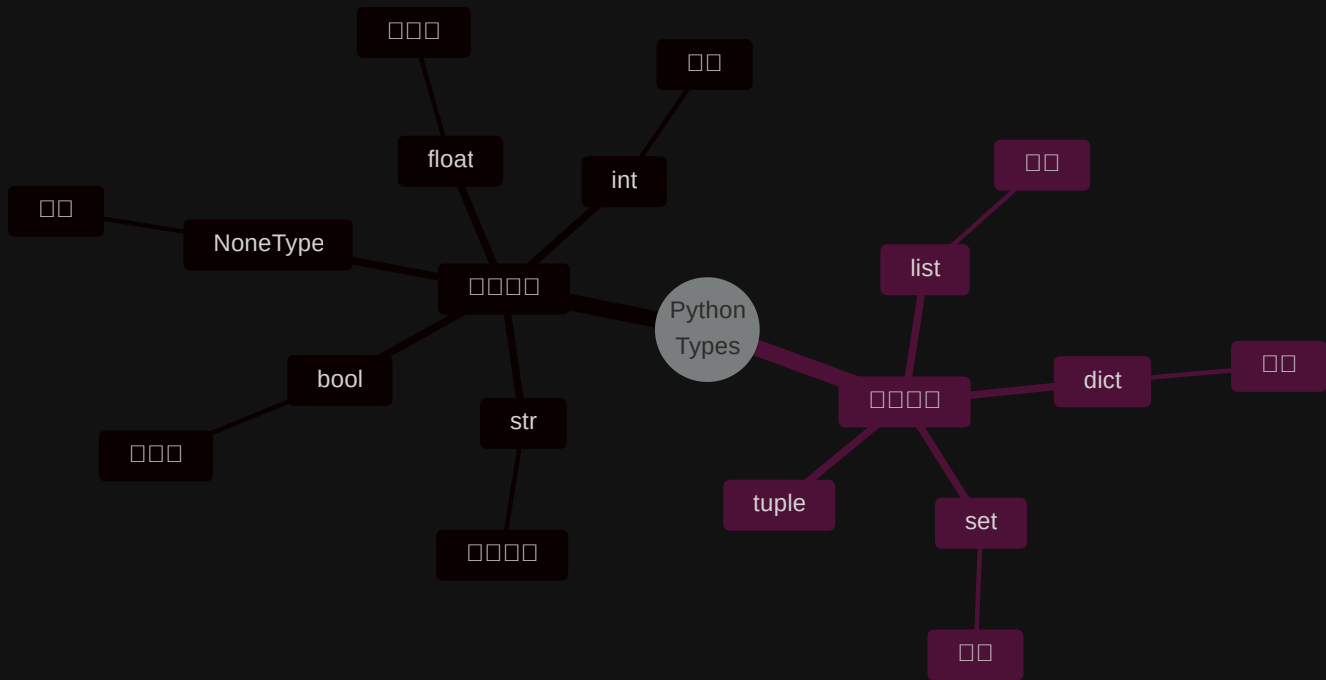
float
□□□

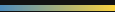
```
str
[]
```

bool

☐ ☐

NoneType
□□





```
# [] [] [] [] []  
'123' ~ [] []  
123 ~ [] []  
'123' == 123 → False
```

'123'



123



```
>>> '123' == 123  
False # [] [] [] [] [] [] [] []
```

 [] [] [] [] []     input() [] [] [] []








































🤔

1. $2x^2 + 3x - 5$
2. $x^2 - 4x + 7$
3. $3x^2 - 2x + 1$
4. $x^2 + 5x - 6$
5. $2x^2 - 7x + 3$
6. $x^2 - 9x + 14$
7. $3x^2 + 4x - 8$
8. $x^2 - 6x + 9$
9. $2x^2 - 5x + 2$
10. $x^2 - 3x + 2$
11. $3x^2 - 7x + 4$
12. $x^2 - 8x + 15$
13. $2x^2 - 9x + 4$
14. $x^2 - 10x + 25$
15. $3x^2 - 11x + 6$
16. $x^2 - 12x + 36$
17. $2x^2 - 13x + 7$
18. $x^2 - 14x + 49$
19. $3x^2 - 15x + 8$
20. $x^2 - 16x + 64$



□ □ □ □ □ □ □ □ □ □ □ □



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! if, for, while



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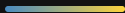
```
1    a, b = 10, 20
2    print(a)    # 10
3    print(b)    # 20
```

✨ □□□□ Python —————

PART 03

Output

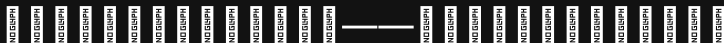
```
❏ print() [] [] [] [] []
```



```
1 print('hi')
2 print('hi')      # 00000000 hi 000
```



```
1 hi
2 hi
```



```
1 # 😞 000 - 000
2 name = 'Lucas'
3 age = 18
4 print('Hello,', name, 'you are', age, 'years old')
```

```
1 # 😊 0 f-string - 0000
2 name = 'Lucas'
3 age = 18
4 print(f'Hello, {name}! You are {age} years old.')
```

```
1 # 🌟 f-string 0000000
2 name = 'Lucas'
3 age = 18
4 print(f'Hello, {name}! Next year you will be {age + 1}')
```



print()



□□□□ □□□□ □□□□ □□□□

```
1 print('hi')
2 print('hi')
3 # :
4 # hi
5 # hi
```

🔑 end — '\n' · sep — ' ' □

f-string

f-string

```
1  # 
2  print(f'{}')
3
4  # 
5  pi = 3.14159265
6  print(f'π ≈ {pi:.2f}')    # π ≈ 3.14
7  print(f'π ≈ {pi:.4f}')    # π ≈ 3.1416
8
9  # 
10 print(f'{{42:05d}}')      # 00042
```

```
{:}
:.2f  - 
:05d  - 
```

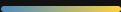


```
1  pi = 3.14159265
2  print(f'π ≈ {pi:.2f}')
3  print(f'π ≈ {pi:.4f}')
4  print(f'{{42:05d}}')
```

PART 04

Input

□□□□□□□□□□



```
sequenceDiagram
    participant P as Person
    participant L as Laptop
    participant M as Monitor
    P->>L: input()
    L->>L: 
    L->>M: print()
    M-.->>P: 
```




`input()` □□□□□□□□□□□□

```
1 n = input()
2 nn = input('□□□□□□□')
```

⚠ `input()` □□□□□□□□□□□□□□□□
! '123' != 123



```
1 # □□□□□□□□□□
2 name = input('□□□□□ ')
3 print(f'Hello, {name}!')
4
5 # input() □□□□□□□□
6 age_str = input('□□□ ')
7 print(type(age_str)) # <class 'str'>
8
9 age = int(age_str) # □□□□
10 print(f'□□□□ {age + 1} □□')
```



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sys.stdin while + try □□□□

```
1  import sys
2
3  for line in sys.stdin:
4      line = line.strip()
5      print(line)
```

 sys.stdin

□□□
□□□□□□

 while+try

□□□□
□□□□□□

 □□□□

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□□□□□□



a.py ☐

□ □ □ □ □ □

□ □ □ □ □

in.txt ☐

```
1 Everything will get better 12345
```



□□□□□□□□

  in.txt □□□□□□□□□□□□□□□□

```
1 import sys
2
3 # UTF-8
4 sys.stdin.reconfigure(encoding='utf-8')
5
6 for line in sys.stdin:
7     line = line.strip()
8     print(line)
```

□□□□□□□□

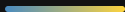
```
1 python a.py < in.txt > out.txt
```

 UTF-8 Windows□□□□□□



□□	□□□	□□	□□□□
□□	<code>n + 2</code>	Addition	<code>n=10 → 12</code>
□□	<code>n - 2</code>	Subtraction	<code>n=10 → 8</code>
□□	<code>n * 2</code>	Multiplication	<code>n=10 → 20</code>
□□	<code>n / 2</code>	Division Not Integer Not Integer Not Integer Not Integer float□	<code>n=10 → 5.0</code>
□□□□	<code>n // 2</code>	□□□□□□□□	<code>n=10 → 5</code>
□□	<code>n ** 6</code>	Power	<code>n=10 → 1000000</code>
□□□	<code>n % 6</code>	Modulo□□□□□□□□□□	<code>n=10 → 4</code>

`int``int``int``/``/``*``*``*``+``+``+``-``-``-``-``-``-`



```
1  n = 10
```

```
n + 2 → 12
```

```
n - 2 → 8
```

```
n * 2 → 20
```

```
n / 2 → 5.0 ← float
```

```
n // 2 → 5
```

```
n ** 6 → 1000000
```

```
n % 6 → 4 ← 10 ÷ 6 商
```

💡 / `float` // `float`

□ □ □ □ □ □ □

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```
1  x = 10
2  x = x + 5    # x 是 15
3  x = x * 2    # x 是 30
4  print(x)
```

⚡ `x += 1` `10` `11` `12` `13` `14` `15` `16` `17` `18` `19` `20` `21` `22` `23` `24` `25` `26` `27` `28` `29` `30` `31` `32` `33` `34` `35` `36` `37` `38` `39` `40` `41` `42` `43` `44` `45` `46` `47` `48` `49` `50` `51` `52` `53` `54` `55` `56` `57` `58` `59` `60` `61` `62` `63` `64` `65` `66` `67` `68` `69` `70` `71` `72` `73` `74` `75` `76` `77` `78` `79` `80` `81` `82` `83` `84` `85` `86` `87` `88` `89` `90` `91` `92` `93` `94` `95` `96` `97` `98` `99` `100`

□□□□□□□□

□□□□□□□□□□

■ ✓ True □□□

■ ✗ False □□□

□□□	□□
<	□□
<=	□□□□
>	□□
>=	□□□□
==	□□
!=	□□□

□□

```
1 10 <= 60      # True
2 123 == 123    # True
3 20 == 40      # False
4 '123' == 123  # False ⚠
```

⚠ '123' 123
False

Logical Operators

Python Logical Operators

Operator	Description
<code>and</code>	Both operands must be true
<code>or</code>	At least one operand must be true
<code>not</code>	Reverses the logical state of its operand

```
1 age = 18
2 has_id = True
3
4 age >= 18 and has_id    # True
5 age < 18 or has_id     # True
6 not has_id              # False
```

🧠 `and` = Both operands must be true · `or` = At least one operand must be true · `not` = Reverses the logical state of its operand

Truthy / Falsy□□□□□□

Python True / False □□□□□□

```
1 bool(0)           # False
2 bool(1)           # True
3 bool("")          # False
4 bool("hi")        # True
5 bool(None)        # False
6 bool([])          # False
```

Falsy False

- 0
- " "
- None
- []
- {}

Truthy True

- 0
-
-

if □□□□

□□□□□□□□

```
1 age = 18
2
3 if age >= 18:
4     print("□□□□")
```

⚠ Python □□□□□□□□□□□□□□□□

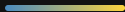
if / elif / else

□□□□□□

```
1  score = 85
2
3  if score >= 90:
4      print("A")
5  elif score >= 80:
6      print("B")
7  elif score >= 70:
8      print("C")
9  else:
10     print("F")
```



if score >= 90: print("A")
elif score >= 80: print("B")
elif score >= 70: print("C")
else: print("F")



```
1 username = input("000")
2 password = input("000")
3
4 if username == "admin" and password == "1234":
5     print("0000")
6 else:
7     print("0000")
```

🔑 000000input + 00 + and 0000

3



- Python □□□□□int / float / str / bool / None

- `input()` □□□□□□□□

- `print()` □□□□□ f-string □□□

- `/` □ `//` □ `%` □□□□□

- □□□□□□□□ + □□□□□□□□

- `if` / `elif` / `else` □□□□

